

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10: GREENHOUSE EFFECT AND CLIMATE CHANGE

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Use everything you have learned in Lessons 1 to 7 to answer the Driving Question: Why are Mount Kenya's glaciers melting and Kenyan farmers' rains becoming unpredictable, and what can the physics of our atmosphere tell us about how to respond? Answer each section in your own words in the space under the prompt. Use evidence from the unit, such as the Lewis Glacier images and data, the testimony of James Muriithi from Embu County, the PhET simulation and the Keeling Curve. Where it helps, draw a labelled diagram. Write in full sentences and use the correct physics words: shortwave radiation, longwave infrared radiation, absorb, re-emit, back-radiation, greenhouse gas, natural greenhouse effect and enhanced greenhouse effect. Link each answer back to Mount Kenya and the long rains.

Part 1: What the Evidence Shows

Look back at the anchoring phenomenon from Lesson 1 and the data you met later in the unit. (a) Describe two pieces of evidence, with numbers, showing that the Lewis Glacier on Mount Kenya has changed. (b) Describe what James Muriithi, the maize farmer in Embu County, reported about the long rains and his harvests. (c) Explain why it is puzzling that a glacier on a mountain and the rains on a farm hundreds of kilometres away are changing at the same time, and what you would need to know to decide whether one cause could be responsible for both.

(a) First, the data strip shows the glacier's area shrinking from about 1.35 km² in 1934 to less than 0.20 km² in 2023. Second, the Lewis Glacier has lost over 90% of its mass since 1900. The paired images from 1934 and today show the same retreat visually. (b) James Muriithi reported that the long rains (March to May) now arrive weeks later than they did in his grandparents' time, and that his maize and bean yields have fallen sharply. (c) It is puzzling because ice melts only when heat is added, yet nothing about the mountain itself has obviously changed, and the two changes happen far apart, one on a cold peak and one on a farm. To decide whether one cause could explain both, we would need to find a process that can change the energy of the whole region, affecting both ice and rainfall. The rest of the unit investigates this.

Part 2: The Physics of the Greenhouse Effect

Explain how the greenhouse effect works. In your answer, trace the path of energy from the Sun to Earth's surface and then back out toward space. Name the type of radiation at each step. Explain what greenhouse gas molecules such as CO₂, water vapour and CH₄ do, and what you saw in the PhET simulation about N₂ molecules compared with CO₂ and CH₄. Then explain the difference between the natural greenhouse effect and the enhanced greenhouse effect.	Incoming shortwave (visible) solar radiation passes through the atmosphere and greenhouse gas molecules without being absorbed. It is absorbed by Earth's surface, which warms. The warm surface then emits the energy as longwave infrared radiation. Greenhouse gas molecules (CO₂, water vapour, CH₄, N₂O) absorb this infrared radiation and re-emit it in all directions, including back toward the surface. This back-radiation warms the surface more than it would be warmed with no atmosphere. In the PhET simulation, N₂ molecules did not absorb infrared photons at all, but CO₂ and CH₄ molecules did. Removing the atmosphere made the surface temperature drop dramatically, and increasing CO₂ and CH₄ raised the equilibrium surface temperature. The natural greenhouse effect is the warming that greenhouse gases cause at natural concentrations, which keeps Earth warm enough for life. The enhanced greenhouse effect happens when human activities raise greenhouse gas concentrations above natural levels, so more infrared radiation is absorbed and re-emitted back to the surface and less energy escapes to space.
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Part 3: From Kenyan Activities to Climate Change

Build a complete cause-and-effect chain that starts with a human activity in Kenya and ends with the retreat of the Lewis Glacier. Name at least two Kenyan activities that increase greenhouse gases and the gas each one releases. Use the Keeling Curve and the global temperature anomaly graph to say what evidence shows that rising greenhouse gases and rising temperature are linked. Say which of these links is shown by correlation in data and which is explained by physics.	Kenyan activities that release greenhouse gases include burning charcoal and clearing forest such as the Mau Forest, which release CO₂, and cattle farming, which releases CH₄ through digestion. Agricultural soils release N₂O. These activities, together with burning petrol and diesel in places like Nairobi, raise greenhouse gas concentrations above natural levels. More greenhouse gas molecules absorb more outgoing infrared radiation and re-emit more of it back toward the surface. This enhanced back-radiation means the surface and the lower atmosphere gain energy and warm. The warmer surface and air around Mount Kenya melt more ice than the glacier gains as new snow, so the Lewis Glacier shrinks and retreats. The Keeling Curve shows atmospheric CO₂ rising from 1958 to the present, and the global temperature anomaly graph shows temperature rising too. The data show a strong upward correlation between the two. The correlation alone does not prove cause, but the physics of infrared absorption by greenhouse gases explains why more CO₂ leads to more warming, so together the data and the mechanism support the claim.
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Part 4: Why the Rains Are Becoming Unpredictable

Use the physics you have learned to explain why the long rains in Embu County have become unpredictable. In your answer, (a) say what happens	(a) When enhanced back-radiation raises surface temperatures, for example over Lake Victoria, evaporation increases. (b) More evaporation changes the moisture and energy in the atmosphere and intensifies the water cycle. This alters the moisture and energy dynamics of the long-rains system, so the March to May rains no longer follow the pattern that farmers have relied on for generations. This is consistent with the delay of 2 to 3 weeks in the onset of the long rains that the Kenya Meteorological Department has recorded since the 1980s. Changes in atmospheric circulation, including the timing of the Inter-Tropical
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to evaporation when surface temperatures rise, (b) explain how this affects the water cycle and the rains over Kenya, and (c) explain why one cause can explain both the shrinking glacier and the farmer's experience. Finally, explain why the ozone layer is not the main cause of either change.	Convergence Zone, are one way scientists link warming to shifts in the seasons. The disrupted planting seasons lower maize and bean yields, as James Muriithi described. (c) Both changes come from the same cause, the enhanced greenhouse effect. Extra energy retained in the Earth system melts ice on the mountain and also disturbs the water cycle that brings rain to farms. The ozone layer is a separate shield in the stratosphere that absorbs ultraviolet (UV-B) radiation. Its damage by CFCs affects UV levels, not the trapping of infrared radiation in the troposphere, so ozone depletion is not the main cause of the glacier retreat or the changing rains. CFCs are also greenhouse gases, so the two problems interact.
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Part 5: Responding to the Problem	
The driving question also asks what the physics can tell us about how to respond. (a) Using your causal chain, suggest two Kenyan actions and explain at which step of the chain each one acts. (b) Explain what the Montreal Protocol shows about how the world can respond to atmospheric damage. (c) Name one question about climate change that you still have after this unit.	(a) Olkaria Geothermal produces electricity without burning fossil fuels, so it reduces the release of CO ₂ at the source of the chain. Energy-efficient household jikos burn less charcoal, so they cut CO ₂ emissions and reduce pressure on forests. Protecting forests such as the Mau also keeps carbon stored in trees. Each action reduces greenhouse gas concentrations, so less infrared radiation is absorbed and sent back to the surface, and warming slows. (b) The Montreal Protocol (1987) reduced CFC production, and the ozone layer is gradually recovering. This shows that when countries act together on clear scientific evidence, atmospheric damage can be reversed, which gives hope that similar action on greenhouse gases could work. (c) Any reasonable open question, for example: How quickly could the Lewis Glacier stabilise if emissions are cut? Or: How can Embu farmers plan planting dates while the long rains are unpredictable?

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the unit	Uses specific data from the lessons accurately, such as the glacier area of about 1.35 km ² in 1934 falling to less than 0.20 km ² in 2023, the loss of over 90% of its mass since 1900, the Embu farmer's testimony, the Keeling Curve and the PhET observations, and links each to a claim.	Uses some evidence from the unit correctly but with limited detail or missing a number, and links most evidence to claims.	Gives little or no evidence, or evidence is inaccurate or not linked to a claim.
Physics of the greenhouse effect	Correctly traces shortwave radiation in, longwave infrared out, absorption and re-emission by greenhouse gases, and back-radiation, and clearly distinguishes the natural from the enhanced	Describes the main steps of the mechanism with minor gaps or imprecise terms, and mentions the natural and enhanced effects.	Describes warming in general terms, with missing steps or confusion between types of radiation.

	greenhouse effect.		
Causal chain and explanation of both phenomena	Builds a complete connected chain from Kenyan human activity to greenhouse gas increase, to enhanced back-radiation, to warming, to glacier loss and altered rainfall, and explains why one cause accounts for both.	Builds a mostly complete chain with one step missing or weakly explained.	Gives isolated statements without a connected chain, or explains only one of the two phenomena.
Distinguishing ideas (ozone, correlation, mechanism)	Clearly separates ozone depletion from the greenhouse effect and the correlation shown in data from the explanation by physics.	Separates these ideas with some imprecision.	Confuses ozone depletion with the greenhouse effect or treats correlation as the explanation.
Responses and communication	Suggests Kenyan responses placed at the correct step of the chain, uses the Montreal Protocol as evidence, and writes clearly with correct scientific terms and a relevant open question.	Suggests reasonable responses with limited link to the chain and mostly clear writing.	Gives vague responses or unclear writing with few scientific terms.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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